Laboratory Report

Application Development Laboratory (CS-33002)

B.Tech Program in ECSC

Submitted by

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Exp No.	TITLE	Date of Experiment	Date of Submission	Remarks
4	To develop an application for user authentication and document sharing	28/01/2026	30/01/2026	

Experiment Number	4
Experiment Title	To develop an application for user authentication and document sharing
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OBJECTIVE:-

To develop an application for user authentication and document sharing.

Detailed Procedure:

1. Install MySQL/MongoDB workbench in your system and install flask-mysqldb package and pymongo. 2. Create a database where you wish to store your user name and the password
3. Implement user authentication/registration form using Flask and the database. For a new user the account is created using the ‘signup’ button. Existing users can directly login with their credentials. 4. Inside the users can update their personal details, reset their passwords. 5. Inside the users can see the grades for their marks, which they cannot edit personally
6. Build a responsive front-end for user interactions.

CODE:-APP.PY:

```
from flask import Flask, render_template, request, redirect, session, flash
from flask_mysqldb import MySQL
from flask_bcrypt import Bcrypt

app = Flask(__name__)
app.secret_key = "secretkey"

app.config['MYSQL_HOST'] = 'localhost'
app.config['MYSQL_USER'] = 'root'
app.config['MYSQL_PASSWORD'] = 'genex206'
app.config['MYSQL_DB'] = 'user_auth'

mysql = MySQL(app)
bcrypt = Bcrypt(app)

@app.route('/')
def home():
    return render_template('login.html')

@app.route('/signup', methods=['POST'])
def signup():
    username = request.form['username']
    email = request.form['email']
    password = bcrypt.generate_password_hash(
        request.form['password']).decode('utf-8')

    cur = mysql.connection.cursor()
    cur.execute(
        "INSERT INTO users(username,email,password) VALUES(%s,%s,%s)",
        (username, email, password))
    mysql.connection.commit()
    cur.close()

    flash("Account created successfully. Please login.", "success")
    return redirect('/')
  -------------------------          -------------------------

@app.route('/login', methods=['POST'])
def login():
    username = request.form['username']
    password = request.form['password']

    cur = mysql.connection.cursor()
    cur.execute("SELECT password FROM users WHERE username=%s", [username])
    user = cur.fetchone()
    cur.close()

    if user and bcrypt.check_password_hash(user[0], password):
        session['user'] = username
```

```
        return jsonify({"error": "Could not extract text"}), 400
    return jsonify({"message": "Document uploaded successfully"})
@app.route("/ask", methods=["POST"])
def ask():
    global DOCUMENT_TEXT
    data = request.get_json()
    question = data.get("question")
    if not DOCUMENT_TEXT:
        return jsonify({"answer": "Please upload a document first."})
    prompt = f"""
Document:
{DOCUMENT_TEXT}
Question:
{question}
Answer clearly and briefly.
"""
    response = llm.complete(prompt)
    return jsonify({"answer": str(response)})
# ------------------------- Run -------------------------
if __name__ == "__main__":
    print("### DOCUMENT CHAT APP RUNNING ###")
    app.run(debug=False, use_reloader=False)
```

CODE:-LOGIN.HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>KIIT Login Page</title>

    <!-- Bootstrap 5 -->
    <link  href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"  rel="stylesheet">
    <style>
        body {
            min-height: 100vh;
            background: linear-gradient(135deg, #0f2027, #203a43, #2c5364);
            display: flex;
            align-items: center;
            justify-content: center;
            font-family: 'Segoe UI', sans-serif;
        }
        .auth-card {
            background: rgba(255, 255, 255, 0.12);
            backdrop-filter: blur(12px);
            border-radius: 18px;
            padding: 35px;
            width: 100%;
            max-width: 430px;
            box-shadow: 0 15px 30px rgba(0,0,0,0.35);
            color: #fff;
        }
        .auth-title {
            text-align: center;
            font-weight: 700;
            letter-spacing: 1px;
        }
        .auth-subtitle {
            text-align: center;
            font-size: 14px;
            opacity: 0.85;
            margin-bottom: 15px;
        }
        .form-control {
            border-radius: 12px;
            padding: 10px;
        }
        .form-control:focus {
            box-shadow: none;
            border-color: #00c6ff;
        }
        .btn-custom {
            background: linear-gradient(135deg, #00c6ff, #0072ff);
```

```
            border: none;
            border-radius: 12px;
            font-weight: 600;
            color: #fff;
            padding: 10px;
        }
        .btn-custom:hover {
            opacity: 0.9;
        }
        .nav-pills .nav-link {
            color: #fff;
            border-radius: 10px;
        }
        .nav-pills .nav-link.active {
            background-color: #fff;
            color: #203a43;
            font-weight: 600;
        }
        footer {
            text-align: center;
            font-size: 12px;
            margin-top: 20px;
            opacity: 0.8;
        }
    </style>
</head>
<body>
<div class="auth-card">
    <!-- Heading -->
    <h3 class="auth-title">KIIT Login Page</h3>
    <p class="auth-subtitle">User Authentication & Academic Portal</p>
    <!-- Flash Messages -->
    {% with messages = get_flashed_messages(with_categories=true) %}
      {% if messages %}
        {% for category, message in messages %}
          <div class="alert alert-{{ category }} text-center">
            {{ message }}
          </div>
        {% endfor %}
      {% endif %}
    {% endwith %}
    <!-- Tabs -->
    <ul class="nav nav-pills mb-4 justify-content-center">
        <li class="nav-item">
            <button class="nav-link active" data-bs-toggle="pill" data-bs-target="#login">
                Login
            </button>
        </li>
        <li class="nav-item">
            <button class="nav-link" data-bs-toggle="pill" data-bs-target="#signup">
                Signup
            </button>
        </li>
    </ul>
    <div class="tab-content">
        <!-- Login Form -->
        <div class="tab-pane fade show active" id="login">
            <form action="/login" method="post">
                <div class="mb-3">
                    <input type="text" name="username" class="form-control"
                           placeholder="Username" required>
                </div>
                <div class="mb-3">
                    <input type="password" name="password" class="form-control"
                           placeholder="Password" required>
                </div>
                <button class="btn btn-custom w-100">Login</button>
            </form>
        </div>
        <!-- Signup Form -->
        <div class="tab-pane fade" id="signup">
            <form action="/signup" method="post">
                <div class="mb-3">
                    <input type="text" name="username" class="form-control"
```

```
                        placeholder="Username" required>
                </div>
                <div class="mb-3">
                    <input type="email" name="email" class="form-control"
                        placeholder="Email" required>
                </div>
                <div class="mb-3">
                    <input type="password" name="password" class="form-control"
                        placeholder="Password" required>
                </div>
                <button class="btn btn-custom w-100">Create Account</button>
            </form>
        </div>
    </div>
    <footer>
        © 2026 Kalinga Institute of Industrial Technology
    </footer>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

CODE:-Dashboard.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>KIIT Student Dashboard</title>

    <!-- Bootstrap 5 -->
    <link  href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"  rel="stylesheet">
    <!-- Icons -->
    <link  href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.1/font/bootstrap-icons.css"  rel="stylesheet">
    <style>
        body {
            background-color: #f4f6f9;
            font-family: 'Segoe UI', sans-serif;
        }
        .sidebar {
            min-height: 100vh;
            background: linear-gradient(180deg, #203a43, #2c5364);
            color: #fff;
            padding: 20px;
        }
        .sidebar h4 {
            font-weight: 700;
            margin-bottom: 30px;
            text-align: center;
        }
        .sidebar a {
            color: #fff;
            text-decoration: none;
            display: block;
            padding: 10px 15px;
            border-radius: 10px;
            margin-bottom: 8px;
            font-weight: 500;
        }
        .sidebar a:hover,
        .sidebar a.active {
            background-color: rgba(255,255,255,0.2);
        }
        .content {
            padding: 30px;
        }
        .card {
            border-radius: 15px;
            box-shadow: 0 8px 20px rgba(0,0,0,0.08);
        }
```

```
        .welcome-box {
            background: linear-gradient(135deg, #00c6ff, #0072ff);
            color: #fff;
            border-radius: 15px;
            padding: 25px;
            margin-bottom: 30px;
        }
        table th {
            background-color: #203a43;
            color: #fff;
        }
    </style>
</head>
<body>
<div class="container-fluid">
    <div class="row">
        <!-- Sidebar -->
        <div class="col-md-3 col-lg-2 sidebar">
            <h4>KIIT Portal</h4>
            <a href="/dashboard" class="active">
    <i class="bi bi-speedometer2"></i> Dashboard
</a>
<a href="/profile">
    <i class="bi bi-person"></i> Profile
</a>
<a href="/reset_password">
    <i class="bi bi-key"></i> Reset Password
</a>
<a href="#">
    <i class="bi bi-bar-chart"></i> Grades
</a>
<a href="/logout" class="text-warning">
    <i class="bi bi-box-arrow-right"></i> Logout
</a>
        </div>
        <!-- Main Content -->
        <div class="col-md-9 col-lg-10 content">
            <!-- Welcome -->
            <div class="welcome-box">
                <h3>Welcome, {{ session['user'] }}</h3>
                <p>Student Dashboard – Academic Overview</p>
            </div>
            <!-- Grades -->
            <div class="card p-4">
                <h5 class="mb-3">Your Grades</h5>
                <table class="table table-bordered table-hover">
                    <tr>
                        <th>Subject</th>
                        <th>Marks</th>
                    </tr>
                    {% for g in grades %}
                    <tr>
                        <td>{{ g[0] }}</td>
                        <td>{{ g[1] }}</td>
                    </tr>
                    {% endfor %}
                </table>
            </div>
        </div>
    </div>
</div>
</body>
</html>
```

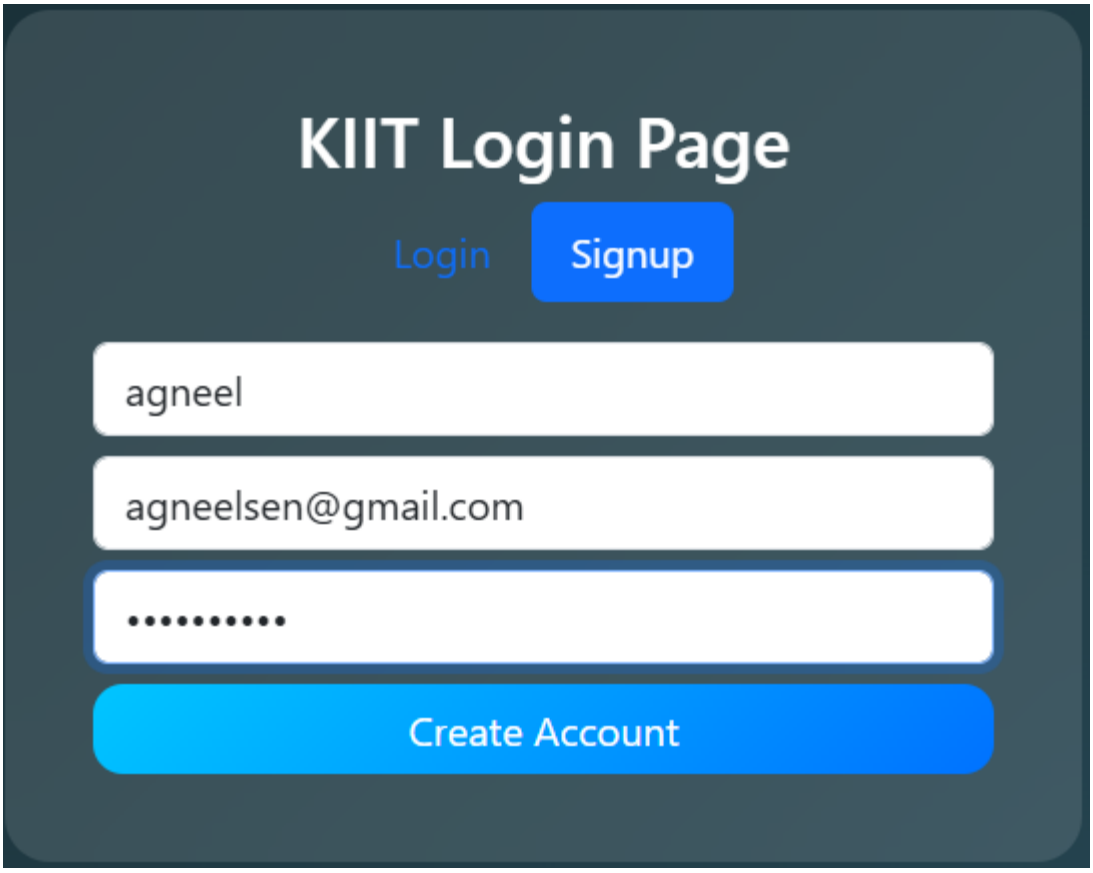
Database:

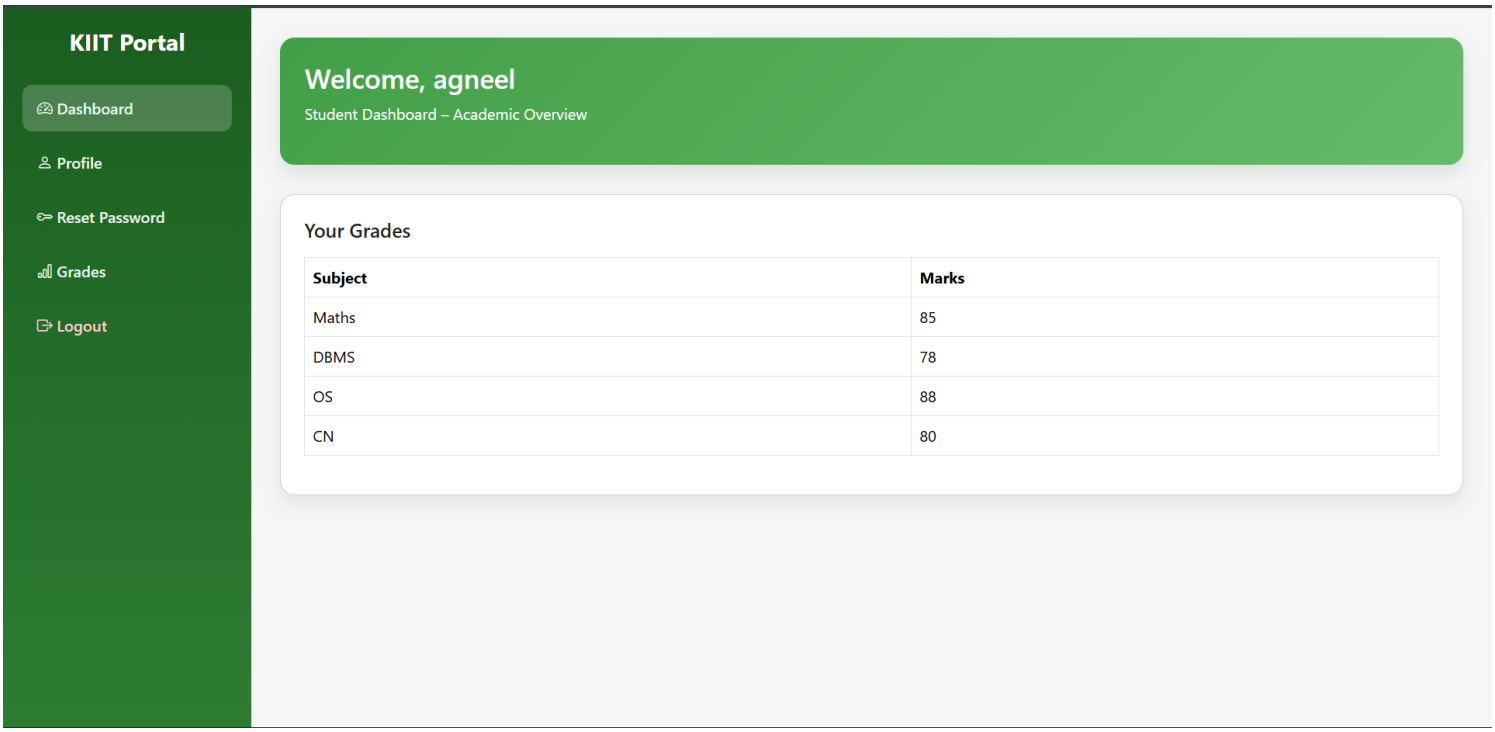
```sql
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50),
    email VARCHAR(100),
    password VARCHAR(255)
);

CREATE TABLE grades (
    id INT AUTO_INCREMENT PRIMARY KEY,
    subject VARCHAR(50),
    marks INT
);

INSERT INTO grades(subject, marks) VALUES
('Maths', 85),
('DBMS', 78),
('OS', 88),
('CN', 80);
```

```sql
    id INT AUTO_INCREMENT PRIMARY KEY,
    subject VARCHAR(50),
    marks INT
);

INSERT INTO grades(subject, marks) VALUES
('Maths', 85),
('DBMS', 78),
('OS', 88),
('CN', 80);

select * from users;
```

id	username	email	password
1	agneel	agneelsen@gmail.com	$2b$12$AA70b6EBJCRmuxAZI4KXx.GmU02YL1...
NULL	NULL	NULL	NULL

OUTPUT:


KIIT Login Page
Login
Signup
agneel
agneelsen@gmail.com
Create Account



REMARKS:-

The experiment was successfully implemented using Flask and MySQL to provide secure user authentication and data management. Passwords were stored in hashed form to ensure security and privacy. The system allowed users to update personal details and reset passwords while restricting grade modification. A responsive Bootstrap-based interface enhanced usability and user interaction.

Agneel Sen
Signature of the Student

Signature of the Faculty